PART 01

# Customer Support Analysis

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## Background

Many customers tweet a brand for help when their products go wrong. In late 2017, support teams faced two very different surges: Apple's iOS 11.1 keyboard bug (released Oct 31, fixed Nov 9) and Amazon's Black Friday (Nov 24) and Cyber Monday (Nov 27).

## Research questions

**RQ1** Does the posting time and words affect the influence the response speed?

- Morning or Night?
- Rude or Polite?

**RQ2** How do support teams react a planned peak vs an unplanned emergency?

- Planned Black Friday vs iOS 11.1 keyboard bug

**RQ3** When does similar-case search help: keywords (TF-IDF) or semantic embeddings?

# Data

We use https://www.kaggle.com/datasets/thoughtvector/customer-support-on-twitter as our data. About 2.8M tweets between customers and big brands' support accounts, mostly Oct–Dec 2017. There are 7 columns:

- tweet_id: an unique id for this tweet.
- author_id: the unique id for the author of the tweet.
- inbound: whether or not the tweet was sent to a company
- created_at: a timestamp of when the tweet was posted
- text: the content of the tweet.
- response_tweet_id: the tweet that responded to this one, if any
- in_response_to_tweet_id: the tweet this tweet was in response to, if any

We mainly focus on the AppleSupport and AmazonHelp in this project.

# Methodology

1. Data Cleaning
   a. Merge the support request and response
   b. Handle missing data
2. EDA (RQ1)
   a. Calculate the response time and analyze the effect of post time
   b. Extract keywords using regular expressions and analyze word.
   c. Data visualization
3. Analysis (RQ2)
   a. Find the the impact of emergency events(Planned and Unplanned) on response performance
4. Modelling (RQ3)
   a. Build TF-IDF and sentence embeddings search system with cosine similarity and top-k.
   b. Analyze which strategy is better in different situations.
5. Prediction
   a. Provide the estimated time of waiting for customers according to the words they use and the post time.
   b. Provide similar cases that other customers asked.

# Expected Results

- Explore how posting time and word relate to response times.
- Compare support performance during Amazon's shopping peaks and Apple's iOS issue.
- Evaluate keyword and semantic search for finding similar cases.
- Build a simple tool to search past cases and estimate waiting times.

## Timeline

- Week 1: Data cleaning and EDA (solve RQ1)
- Week 2: Analysis the impact of planned peak and emergency (solve RQ2)
- Week 3: Analysis the methods of searching and build a recommendation system (solve RQ3)
- Week 4: Prepare for the presentation (slides or jupyter notebook)